

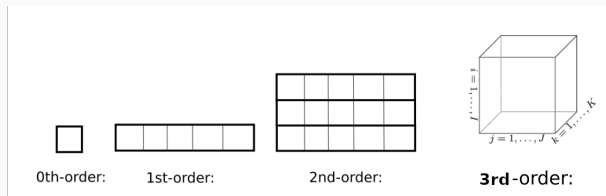
Mixed Precision Iterative Refinement for Low-Rank Matrix and Tensor Approximations

Matthieu Robeyns

Joint work with Marc Baboulin, Oguz Kaya, and Theo Mary

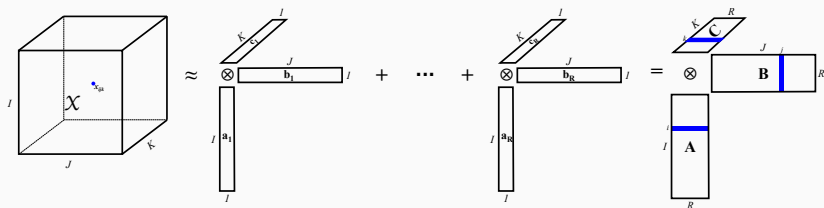
SIAM CSE 2023, March 2, 2023

Tensor definition



- A **tensor**, denoted as $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_d}$, is a multidimensional array of data.
- The **order** of the tensor is its number of dimensions.
- Powerful tool in many applications : data analysis, quantum computing, AI, signal processing, scientific computing, ...
- The size of a tensor is exponential in its order $\Rightarrow$ **curse of dimensionality**.

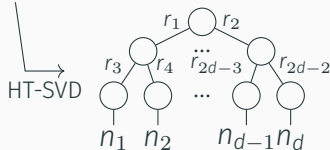
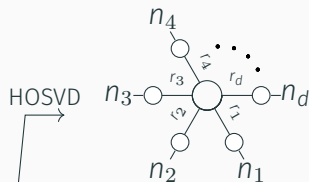
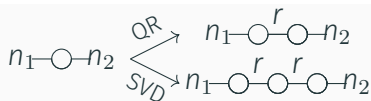
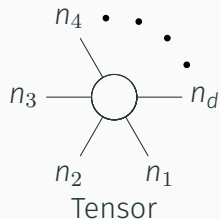
Low-rank approximation



Rank r CP decomposition of $\mathcal{X}$

- **Low-rank approximation (LRA)** is the problem of approximating an order d tensor by smaller tensors with **inner dimensions** as the **rank** of decomposition $\Rightarrow$ **curse of dimensionality solved**.

Different types of low-rank decompositions



Direct methods

Solve the LRA problem using a fixed error threshold, and give a quasi-optimal approximation.

Iterative methods

Solve the LRA problem with a sequence of approximations, each approximation improving upon the previous one. No guarantee of convergence.

Low-rank approximation

Direct methods

Solve the LRA problem using a fixed error threshold, and give a quasi-optimal approximation.

Iterative methods

Solve the LRA problem with a sequence of approximations, each approximation improving upon the previous one. No guarantee of convergence.

- Low-rank approximation of tensors heavily uses linear algebra operations.
Computationally intensive.

Accelerating tensor computations

⇒ Need HPC methods (parallelism, accelerators) to speed up computations.

		Range	Precision	Speed (Tflop/s)*
fp64	double	$10^{\pm 308}$	1×10^{-16}	32
fp32	single	$10^{\pm 38}$	6×10^{-8}	64
fp16	half	$10^{\pm 5}$	5×10^{-4}	1000
bfloat16		$10^{\pm 38}$	4×10^{-3}	
fp8 (e4m3)	quarter	$10^{\pm 2}$	6×10^{-2}	2000
fp8 (e5m2)		$10^{\pm 5}$	1×10^{-1}	

*on NVIDIA Hopper H100 SXM5

⇒ **Low precision** computations achieve very high performance with resulting low accuracy.

Mixed Precision algorithm

- Many LRA applications need an accuracy higher than 16-bit.
- We will use **mixed precision** algorithms with the hope of achieving both high performance and high accuracy.

Proposed approach : **Iterative refinement**

1. Apply method to original input in **low** precision.
2. Compute the residual in **high** precision.
3. Apply method to the residual in **low** precision.
4. Combine (1) and (3) to obtain an approximation in **high** precision.

Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

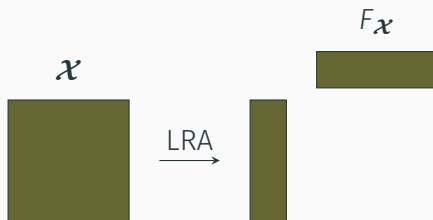
LRA : Some LRA method

1 : $F\mathcal{X} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ

2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F\mathcal{X})$

3 : $F\Delta\mathcal{X} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ

4 : $F'\mathcal{X} \leftarrow F\mathcal{X} + F\Delta\mathcal{X}$



Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

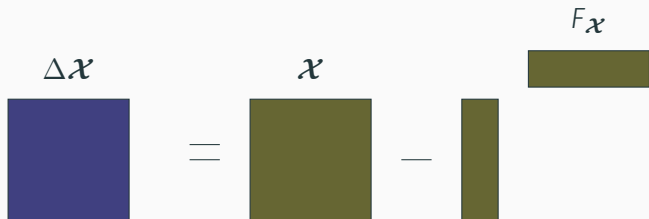
LRA : Some LRA method

1 : $F\mathcal{X} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ

2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F\mathcal{X})$

3 : $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ

4 : $F'_{\mathcal{X}} \leftarrow F\mathcal{X} + F_{\Delta\mathcal{X}}$



Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

1 : $F\mathcal{X} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ

2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F\mathcal{X})$

3 : $F\Delta\mathcal{X} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ

4 : $F'\mathcal{X} \leftarrow F\mathcal{X} + F\Delta\mathcal{X}$

$\Delta\mathcal{X}$



LRA
→



$F\Delta\mathcal{X}$



Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

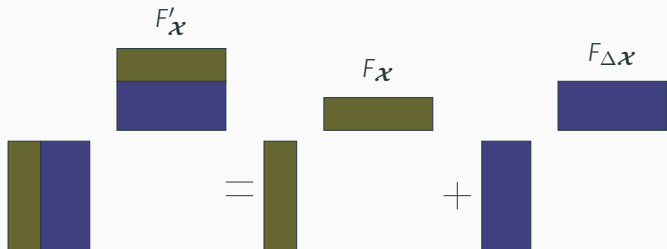
LRA : Some LRA method

1 : $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ

2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$

3 : $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ

4 : $F'_{\mathcal{X}} \leftarrow F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$



Iterative refinement

Input :

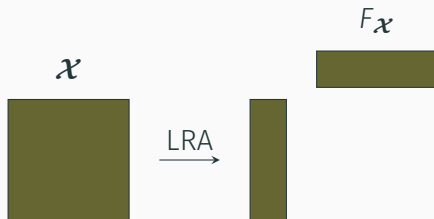
$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

1 : $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ
2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$
3 : $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ
4 : $F'_{\mathcal{X}} \leftarrow F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$

Rank analysis :

- $\text{rank}(\mathcal{X}) \leq \vec{r}$.
- $\text{rank}(F_{\mathcal{X}}) \leq \vec{r}$.



Error analysis :

- $\|\mathcal{X} - F_{\mathcal{X}}\| \leq c_1 u_\ell \|\mathcal{X}\|$.

Where u_ℓ is the low precision unit roundoff.

Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

-
- 1 : $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_{ℓ}
 - 2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$
 - 3 : $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_{ℓ}
 - 4 : $F'_{\mathcal{X}} \leftarrow F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$

$$\text{rank}(\Delta\mathcal{X}) \leq \text{rank}(\mathcal{X}) + \text{rank}(F_{\mathcal{X}})$$

Rank analysis :

- $\text{rank}(\Delta\mathcal{X}) \leq \text{rank}(\mathcal{X}) + \text{rank}(F_{\mathcal{X}})$.
- $\Rightarrow \text{rank}(\Delta\mathcal{X}) \leq 2\vec{r}$.

Error analysis :

- $\|F_{\mathcal{X}} - \tilde{F}_{\mathcal{X}}\| \leq c_2 u \|F_{\mathcal{X}}\|$, where $\tilde{F}_{\mathcal{X}} = \text{DECOMPRESS}(F_{\mathcal{X}})$, and u is the high precision unit roundoff.

Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

1: $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ

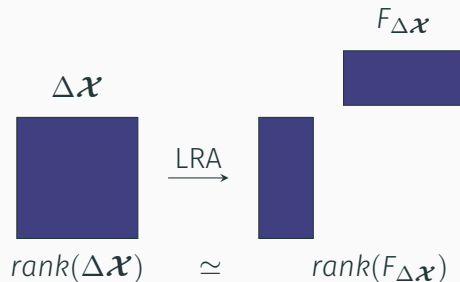
2: $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$

3: $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ

4: $F'_{\mathcal{X}} \leftarrow F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$

Rank analysis :

- $\text{rank}(F_{\Delta\mathcal{X}}) \leq 2\tilde{r}$.



Error analysis :

- $\|\Delta\mathcal{X} - F_{\Delta\mathcal{X}}\| \leq c_1 u_\ell \|\Delta\mathcal{X}\|$.

Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

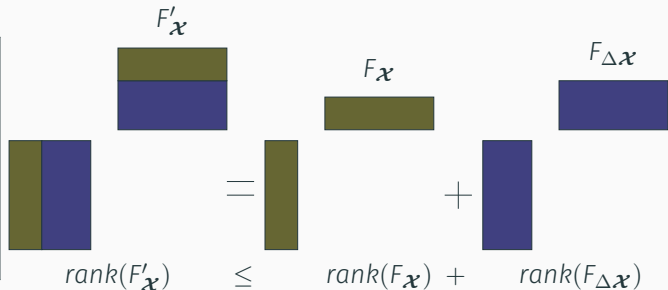
LRA : Some LRA method

1 : $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ $\triangleright$ In low precision u_ℓ

2 : $\Delta\mathcal{X} \leftarrow \mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$

3 : $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ $\triangleright$ In low precision u_ℓ

4 : $F'_{\mathcal{X}} \leftarrow F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$



Rank analysis :

$$\Rightarrow \text{rank}(F'_{\mathcal{X}}) \leq 3\vec{r}.$$

Error analysis :

$$\Rightarrow \|\mathcal{X} - F'_{\mathcal{X}}\| \leq ((c_1 u_\ell)^2 + c_2 u) \|\mathcal{X}\|.$$

Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

n_{iter} : Number of iterations

1: $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$

2: $\Delta\mathcal{X} \leftarrow \mathcal{X}$

3: for $i = 1, \dots, n_{\text{iter}}$ do

4: $\Delta\mathcal{X} \leftarrow \Delta\mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$

5: $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$

6: $F_{\mathcal{X}} \leftarrow F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$

7: end for

Theorem

After i iterations, the computed $F_{\mathcal{X}}$ satisfies

$$\|\mathcal{X} - F_{\mathcal{X}}\| \leq (\phi^{(i+1)} + \xi + O(u_{\ell}u)) \|\mathcal{X}\|$$

- $\phi = O(u_{\ell})$ is the convergence speed
- $\xi = O(u)$ is the attainable accuracy

Rank analysis : $\text{rank}(F_{\mathcal{X}}) \leq \sum_{j=1}^i 2^j \vec{r}$.

Iterative refinement

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

n_{iter} : Number of iterations

```
1:  $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ 
2:  $\Delta\mathcal{X} \leftarrow \mathcal{X}$ 
3: for  $i = 1, \dots, n_{\text{iter}}$  do
4:    $\Delta\mathcal{X} \leftarrow \Delta\mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$ 
5:    $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ 
6:    $F_{\mathcal{X}} \leftarrow \text{RECOMPRESS}(F_{\mathcal{X}} + F_{\Delta\mathcal{X}}, \phi^{(i+1)})$ 
7: end for
```

Theorem

After i iterations, the computed $F_{\mathcal{X}}$ satisfies

$$\|\mathcal{X} - F_{\mathcal{X}}\| \leq (\phi^{(i+1)} + \xi + O(u_{\ell}u)) \|\mathcal{X}\|$$

- $\phi = O(u_{\ell})$ is the convergence speed
- $\xi = O(u)$ is the attainable accuracy

Rank analysis : $\text{rank}(F_{\mathcal{X}}) \leq \vec{r}$.

- $\text{RECOMPRESS}(F, \varepsilon)$: Tighten the ranks of a decomposition F with non-optimal ranks, by finding a lower rank decomposition within its ε vicinity.
 $\Rightarrow$ Should recompress $F_{\mathcal{X}} + F_{\Delta\mathcal{X}}$ back to rank $\vec{r}$.

Cost Analysis

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

n_{iter} : Number of iterations

```
1:  $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$ 
2:  $\Delta\mathcal{X} \leftarrow \mathcal{X}$ 
3: for  $i = 1, \dots, n_{\text{iter}}$  do
4:    $\Delta\mathcal{X} \leftarrow \Delta\mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$ 
5:    $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$ 
6:    $F_{\mathcal{X}} \leftarrow \text{RECOMPRESS}(F_{\mathcal{X}} + F_{\Delta\mathcal{X}}, \phi^{(i+1)})$ 
7: end for
```

Let $P = \prod \text{dims}$ and $S = \sum \text{dims}$.

Let r_i be the rank of $F_{\mathcal{X}}$ at iteration i , with $r_i \leq r$ (final rank).

- $\text{cost}(\text{LRA}) = c_1 P \sum r_i$ flops ($c_1 \simeq [4 : 10]$).
- $\text{cost}(\text{Decompress}) = c_2 P \sum r_i$ flops ($c_2 = 2$).
- $\text{cost}(\text{Recompress}) = O(S \sum r_i)$ flops.

$$\text{cost}(\text{LRA}) \geq \text{cost}(\text{Decompress}) \gg \text{cost}(\text{Recompress})$$



low precision



high precision

Input :

$\mathcal{X}$: Tensor or matrix

LRA : Some LRA method

n_{iter} : Number of iterations

1 : $F_{\mathcal{X}} \leftarrow \text{LRA}(\mathcal{X})$

2 : $\Delta\mathcal{X} \leftarrow \mathcal{X}$

3 : for $i = 1, \dots, n_{\text{iter}}$ do

4 : $\Delta\mathcal{X} \leftarrow \Delta\mathcal{X} - \text{Decompress}(F_{\mathcal{X}})$

5 : $F_{\Delta\mathcal{X}} \leftarrow \text{LRA}(\Delta\mathcal{X})$

6 : $F_{\mathcal{X}} \leftarrow \text{RECOMPRESS}(F_{\mathcal{X}} + F_{\Delta\mathcal{X}}, \phi^{(i+1)})$

7 : end for

Two scenarios where the proposed IR can be much faster :

1. Hardware with **much faster** (x3) low precision arithmetic and/or mixed precision FMA (e.g., GPU tensor cores)
2. $r_i \ll r$ for small i (rapid decay of singular values).

- Code developed in MATLAB(2019a) with the following decompositions :
 - Randomized SVD (Matrix), [Martinsson & Voronin, 2016],
 - QRCP (Matrix),
 - HOSVD (Tucker) [De Lathauwer & al., 2000] TENSOR TOOLBOX,
 - TTSVD (Tensor-Train) [Oseledets & al., 2011] TT-TOOLBOX,
 - HTSVD (Hierarchical-Tensor) [Grasedyck, 2010] HTUCKER_1.2.

CHOP : to simulate low precision.

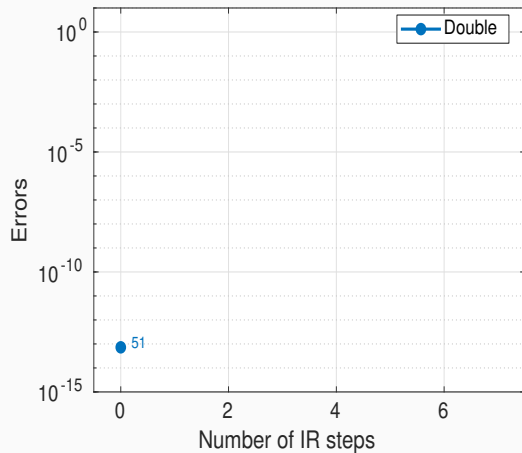
Experimental Setting

- Code developed in MATLAB(2019a) with the following decompositions :
 - Randomized SVD (Matrix), [Martinsson & Voronin, 2016],
 - QRCP (Matrix),
 - HOSVD (Tucker) [De Lathauwer & al., 2000] TENSOR TOOLBOX,
 - TTSVD (Tensor-Train) [Oseledets & al., 2011] TT-TOOLBOX,
 - HTSVD (Hierarchical-Tensor) [Grasedyck, 2010] HTUCKER_1.2.

CHOP : to simulate low precision.

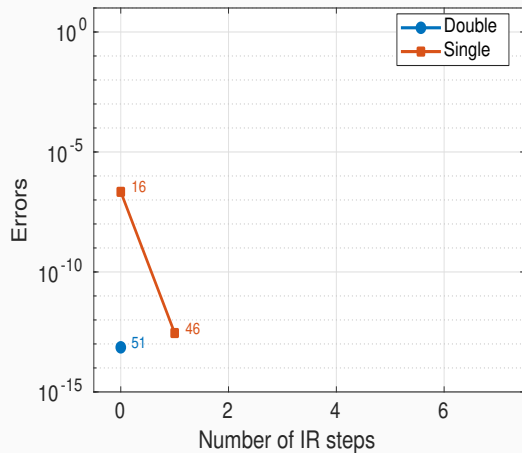
- Our benchmarks involve small matrices ($\simeq 300 \times 300$) and tensors ($\simeq 100 \times 100 \times 100$) randomly generated with a specific decreasing behavior of singular values.

Experiments on matrices



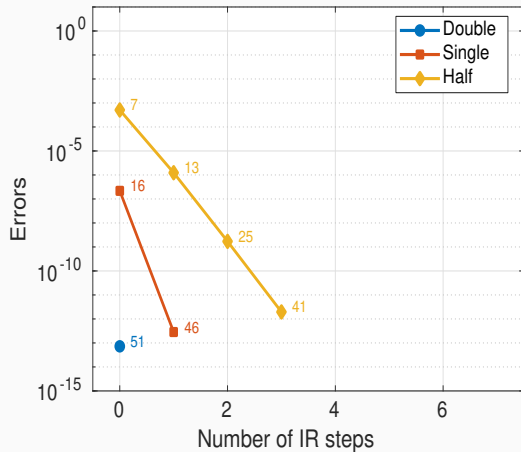
(a) RandSVD computation with a Poisson matrix of size 253×252

Experiments on matrices



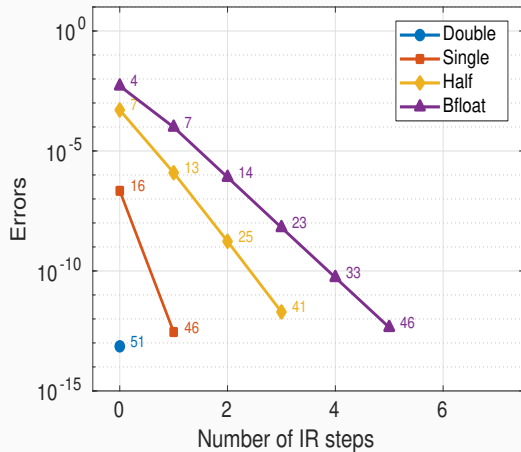
(a) RandSVD computation with a Poisson matrix of size 253x252

Experiments on matrices



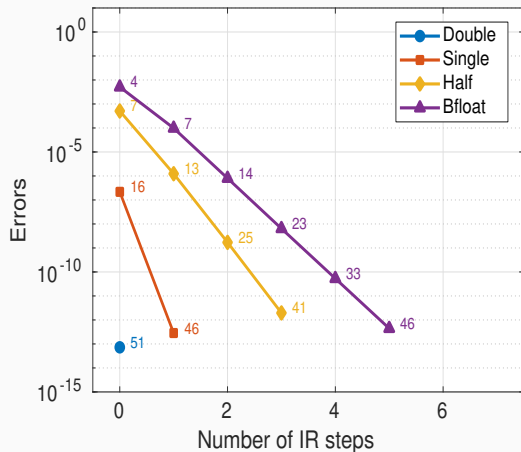
(a) RandSVD computation with a Poisson matrix of size 253x252

Experiments on matrices

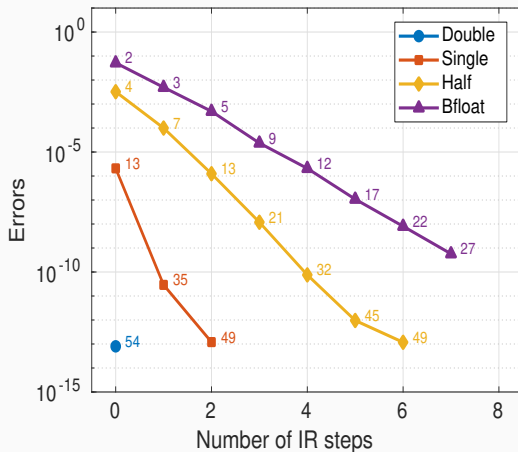


(a) RandSVD computation with a Poisson matrix of size 253x252

Experiments on matrices

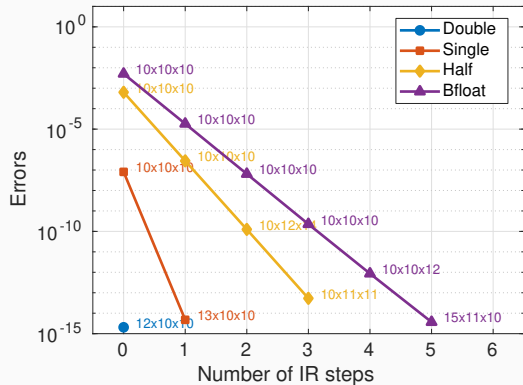


(a) RandSVD computation with a Poisson matrix of size 253x252

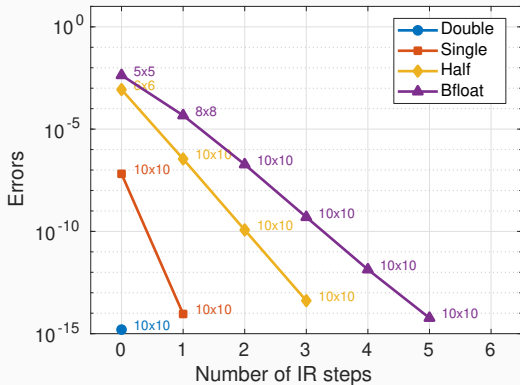


(b) QRCP with the same matrix

Experiments on tensors

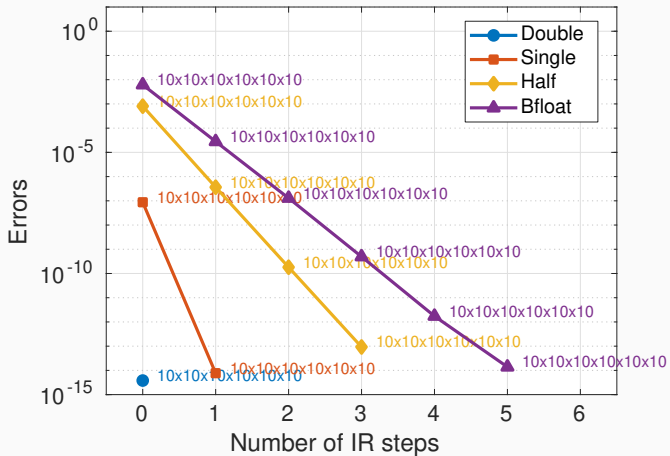


(a) HOSVD with $\vec{r} = 10 \times 10 \times 10$



(b) TT-SVD with $\vec{r} = 10 \times 10$

Experiments on tensors



(a) HT-SVD with
 $\vec{r} = 10 \times 10 \times 10 \times 10 \times 10 \times 10$

Contributions

- We propose a **new** iterative refinement method for tensor LRA.
- We achieve **high** accuracy with most operations in **low** precision.

Future work

- Test the method on real-life tensor benchmarks.
- Develop a parallel high performance implementation of the method with performance analysis, e.g., on GPUs.

Preprint in preparation : M. Baboulin, O. Kaya, T. Mary, M. Robeyns,
Mixed precision iterative refinement for low-rank matrix and tensor approximations.

THANK YOU! Questions?